Quasi-Categories vs. Simplicial Categories

William Balderrama

2/8/2017

The goal of this talk is to show how we can move from one model of higher categories (quasi-categories) to another model of higher categories (simplicial cateories) and also why we might do so.

Motivation

There is the idea that an $(\infty, 1)$-category is like a category, but with higher morphisms that are all invertible. In other words, an $(\infty, 1)$-category is a "category enriched in $(\infty, 0)$-categories". If we take "$(\infty, 0)$-category" = "space", we arrive at:

- Kan complex-enriched categories are a model for $(\infty, 1)$-categories.

The problem with this is that there are natural constructions which only give rise to $(\infty, 1)$-categories with composition that is associative only up to coherent homotopy. In order to capture the desired flexibility, people arrived at:

- Quasicategories are a model for $(\infty, 1)$-categories.

In order to reconcile these two ideas, we need a way of comparing the two models. They do turn out to be equivalent in a precise sense; in particular, every quasicategory can be "rigidified" into a Kan complex-enriched category.

Relevant Model Structures

We can make the comparison between quasicategories and Kan complex-enriched categories precise with the use of model category theory. To that end, we need relevant model categories. On one hand, recall the following.

Theorem 2.1. *There is a model structure on* sSet*, with*

- *Fibrant objects: quasicategories;*

- *Weak equivalences: categorical equivalences.*

We spent a whole talk discussing quasi-categories and how they give rise to suitable notion of "higher category theory", thus for more details see [Pa17].

On the other hand, we have the following.

Theorem 2.2. *There is a model structure on* sCat*, the category of simplicial categories, commonly known as the Bergner model structure, with*

- *Fibrant objects: Kan complex-enriched categories;*
- *Weak equivalences: functors $f\colon \mathcal{C} \to \mathcal{D}$ which are*
  - *(i) Essentially surjective: the underlying functor of* 1*-categories is essentially surjective;*
  - *(ii) Fully faithful: for any $a, b \in \mathcal{C}$, the map $\mathcal{C}(a,b) \to \mathcal{D}(fx, fy)$ is a weak equivalence.*

Remark 2.3. Because of the way fibrant objects are defined we often say: Fibrant objects in the Bergner model structure are the "locally fibrant objects".

We did not spend much time discussing the Bergner model structure. A great source for that would be the original source, namely [Be07]

The comparison is then of the form

Theorem 2.4. *There is a Quillen equivalence*

$$(\mathrm{sSet})^{Joy} \underset{N_\Delta}{\overset{\mathfrak{C}}{\rightleftarrows}} (s\mathrm{Cat})^{Berg}$$

The right adjoint is commonly called the *simplicial nerve functor*, whereas the left adjoint is sometimes called the *rigidification functor*.

This Quillen equivalence was proven in several forms by many people. The one we stated above is the most famous version and was stated and proven by Jacob Lurie in [Lu09] (Theorem 2.2.5.1).

Defining an Adjunction between Presentable Categories

To define our adjunction $\mathfrak{C} : \mathrm{sSet} \rightleftarrows \mathrm{sCat} : N_\Delta$, it suffices to define a functor $\mathfrak{C}\colon \Delta \to \mathrm{sCat}$. Indeed, left adjoints have to commute with colimits and every simplicial set is canonically the colimit of Δ^n (categories which satisfy such condition are generally called *presentable*). Thus, given any simplicial set X, we can write $X = \mathrm{colim}_{(\Delta\downarrow X)}\Delta^n$, and so are forced to define

$$\mathfrak{C}(X) = \mathrm{colim}_{(\Delta\downarrow X)}\mathfrak{C}(\Delta^n).$$

Moreover, if a right adjoint N_Δ exists, then for any simplicial category $\mathcal{C}$ we necessarily have

$$(N_\Delta \mathcal{C})_n = \mathrm{Hom}_{\mathrm{sSet}}(\Delta^n, N_\Delta \mathcal{C}) \cong \mathrm{Hom}_{\mathrm{sCat}}(\mathfrak{C}(\Delta^n), X).$$

Taking this as a definition defines a right adjoint, no matter what $\mathfrak{C}$ we started with.

The left Adjoint

In this section we will focus on defining $\mathfrak{C}$ for generators Δ^n. A suitable functor $\mathfrak{C}\colon \Delta \to \mathrm{sCat}$ can be constructed with the following idea.

- The objects of $\mathfrak{C}(\Delta^n)$ are the vertices $\{0, \dots, n\}$ of Δ^n;
- For $i, j \in \{0, \dots, n\}$, the mapping space $\mathfrak{C}(\Delta^n)(i, j)$ encodes all order-preserving paths from i to j and all order-preserving homotopies between such paths. Concretely, let $P_{i,j}$ be the poset of all subsets of the interval $[i, j]$, that include the end points $\{i, j\}$. Then we have

$$\mathfrak{C}(\Delta^n)(i, j) = N(P_{i,j})$$

For example, to see what $\mathfrak{C}(\Delta^2)$ shoud be, we can draw the picture

$$\begin{tikzcd} & 1 \arrow[dr] \arrow[d, phantom, "\Uparrow"] & \\ 0 \arrow[ur] \arrow[rr] & {} & 2 \end{tikzcd} .$$

So we would have for instance $\mathfrak{C}(\Delta^2)(0, 2)_0 = \{\{0, 2\}, \{0, 1, 2\}\}$. More generally $\mathfrak{C}(\Delta^2)(0, 2) = \Delta^1$, where one vertex represents a map from 0 to 2, the other vertex represents the composition and the unique arrow gives us a unique homotopy from the composition to the map.

A complete definition of $\mathfrak{C}$ can be given along these lines, resulting in the stated Quillen equivalence. However, there is still a major problem. Namely, say we have defined $\mathfrak{C}\colon \Delta \to \mathrm{sCat}$. Then for X a simplicial set, or quasicategory if one prefers, we can check from the formulas that $\mathfrak{C}(X) = \mathrm{colim}_{(\Delta^n \downarrow X)} \mathfrak{C}(\Delta^n)$ has as objects the vertices of X. However, for $a, b \in X_0$, it can be extremely difficult to give a good description of the mapping space $\mathfrak{C}(X)(a, b)$.

That is because there is no simple way to compute colimits in the category of categories, let alone category of simplicial categories.

Example 4.1. As an exercise compute following pushout diagram in the category of categories:

$$\begin{tikzcd} {[0]\coprod[0]} \arrow[r] \arrow[d] & {[1]} \\ {[0]} & \end{tikzcd}$$

Furtunately, there is a reasonable solution to this problem. We can define a new functor $\mathfrak{C}^{\mathcal{N}} : \mathrm{sSet} \to \mathrm{sCat}$, so that

- For every simplicial set X, $\mathfrak{C}^{\mathcal{N}}(X)$ is weakly equivalent to $\mathfrak{C}(X)$;
- It is easier to describe and deal with the mapping spaces in $\mathfrak{C}^{\mathcal{N}}(X)$.

Defining the functor $\mathfrak{C}^{\mathcal{N}}$ is not difficult. However, we will need some preliminary definitions.

From here on everything that is discussed comes from an excellent paper by Dugger and Spivak [DS11], which sets up the theory of necklaces necessary to understand the mapping spaces of the simplicial categories we build out of simplicial sets.

Necklaces

Definition 5.1. A **necklace** is a simplicial set of the form

$$T = \Delta^{n_0} \vee \cdots \vee \Delta^{n_k},$$

where we attach the terminal vertex of Δ^{n_j} to the initial vertex of $\Delta^{n_{j+1}}$.

Notice that if $n_0 = \cdots = n_k = 1$, then T is just a spine. So in general, a necklace can just be thought of as a thickened version of a spine.

Example 5.2. We can draw a picture of the necklace $\Delta^1 \vee \Delta^3 \vee \Delta^2$ as

$$\begin{tikzcd} & & \bullet \arrow[dd] \arrow[dr] & & & \bullet \arrow[dr] & \\ \alpha \arrow[r] & \bullet \arrow[ur] \arrow[dr] \arrow[rrr] & & \bullet \arrow[ur] \arrow[rr] & & & \omega \\ & & \bullet \arrow[ur] & & & & \end{tikzcd}$$

Every necklace is bipointed, with initial vertex α and the terminal vertex ω. So, a necklace T naturally lives in the category $\mathrm{sSet}_{*,*} = (\partial\Delta^1 \downarrow \mathrm{sSet})$ of bipointed simplicial sets. Let $\mathcal{N} \subset \mathrm{sSet}_{*,*}$ denote the resulting category of necklaces.

It is also convenient to introduce the following notation. An object of $\mathrm{sSet}_{*,*}$ is determined by a simplicial set X and two vertices $a, b \in X$; so, we will denote such an object by $X_{a,b}$.

For a more detailed analysis of necklaces and their interesting properties see Section 3 of [DS11].

The Improved Rigidification

If $\mathcal{C}$ is a 1-category and a, b are objects of $\mathcal{C}$, then an element of $\mathrm{Hom}_{\mathcal{C}}(a, b)$ is drawn as

$$a \longrightarrow b \ .$$

If we also want to picture composition in $\mathcal{C}$, which can be considered as maps

$$\mathrm{Hom}(a, c_1) \times \mathrm{Hom}(c_1, c_2) \times \cdots \times \mathrm{Hom}(c_n, b) \to \mathrm{Hom}(a, b)$$

for $c_1, \ldots, c_n \in \mathcal{C}$, then we must allow our pictures of $\mathrm{Hom}_{\mathcal{C}}(a, b)$ to include things like

$$a \longrightarrow c_1 \longrightarrow \cdots \longrightarrow c_n \longrightarrow b \ .$$

This is exactly the image of a spine in $\mathcal{C}$.

By thickening all of this up, we arrive at the following definition.

Definition 6.1. For a simplicial set X, define $\mathfrak{C}^{\mathcal{N}}(X)$ to be the simplicial category with

- Objects: the vertices of X;
- Morphisms: For $a, b \in X$,

  $$\mathfrak{C}^{\mathcal{N}}(X)(a, b) = N(\mathcal{N} \downarrow X_{a,b}),$$

  where N is the ordinary nerve functor;
- Composition: induced by the functor $(\mathcal{N} \downarrow X_{a,b}) \times (\mathcal{N} \downarrow X_{b,c}) \to (\mathcal{N} \downarrow X_{a,c})$ given by

  $$(T \to X_{a,b}, S \to X_{b,c}) \mapsto (T \vee S \to X_{a,c}).$$

As previously mentioned, we have the following.

Theorem 6.2. As a simplicial category, $\mathfrak{C}^{\mathcal{N}}(X)$ is weakly equivalent to $\mathfrak{C}(X)$.

Remark 6.3. Note this weak equivalence is not direct. Rather there is a third functor and a natural zig-zag between these two definitions. For the exact constructions and a proof see section 5 of [DS11].

An application to rigidification of products

Because the mapping spaces of $\mathfrak{C}^{\mathcal{N}}(X)$ are given as nerves of explicitly described ordinary categories, there are more tools available for studying them. To give an example, it is useful to generalize this construction.

Definition 7.1. For a subcategory $\mathcal{P}$ of $\mathrm{sSet}_{*,*}$, simplicial set X, and vertices $a, b \in X$, define

$$\mathfrak{C}^{\mathcal{P}}(X)(a,b) = N(\mathcal{P} \downarrow X_{a,b}).$$

So long as $\mathcal{P}$ contains Δ^0 and is closed under the appropriate wedge sums, $\mathfrak{C}^{\mathcal{P}}(X)$ can be given the structure of a simplicial category.

There are conditions that can be placed on a category $\mathcal{P}$ with $\mathcal{N} \subset \mathcal{P} \subset \mathrm{sSet}_{*,*}$ to ensure that $\mathfrak{C}^{\mathcal{P}}(X)$ is a model for the rigidification of X. In particular, we have the following.

Fact 7.2. Let $\mathcal{P} \subset \mathrm{sSet}_{*,*}$ be the full subcategory of finite products of necklaces. Then for any simplicial set X, the map $\mathfrak{C}^{\mathcal{N}}(X) \to \mathfrak{C}^{\mathcal{P}}(X)$ is a weak equivalence.

Using this, we can give an easy proof of the following.

Proposition 7.3. For any simplicial sets X, Y, the canonical map $\mathfrak{C}(X \times Y) \to \mathfrak{C}(X) \times \mathfrak{C}(Y)$ is a weak equivalence.

Proof. One can check that this map is a bijection on objects, so it suffices to show that for $x_0, x_1 \in X$ and $y_0, y_1 \in Y$, the canonical map

$$\mathfrak{C}^{\mathcal{P}}(X \times Y)((x_0, y_0), (x_1, y_1)) \to \mathfrak{C}^{\mathcal{P}}(X)(x_0, x_1) \times \mathfrak{C}^{\mathcal{P}}(X)(y_0, y_1)$$

is a weak equivalence. To that end, it suffices to give functors

$$f : (\mathcal{P} \downarrow (X \times Y)_{(x_0,y_0),(x_1,y_1)} \rightleftarrows (\mathcal{P} \downarrow X)_{(x_0,x_1)} \times (\mathcal{P} \downarrow X)_{(y_0,y_1)} : g$$

along with natural transformations $1 \Rightarrow gf$ and $fg \Rightarrow 1$. But this is easy; we can simply define

$$\begin{aligned} f &: (T \to X \times Y) \mapsto (T \to X, T \to Y) \\ g &: (T \to X, S \to Y) \mapsto (T \times S \to X \times Y). \end{aligned}$$

Then the desired natural transformations are given componentwise by

$$\begin{aligned} 1 \Rightarrow gf &: (T \to X \times Y) \xrightarrow{\Delta} (T \times T \to X \times Y) \\ fg \Rightarrow 1 &: (T \times S \to X, T \times S \to Y) \xrightarrow{(\pi_T, \pi_S)} (T \to X, S \to Y). \end{aligned}$$

□

Remark 7.4. Note the same result can be found in [Lu09] (Corollary 2.2.5.6), but only after the equivalence between quasi-categories and simplicial categories has already been proven. Thus this simple application shows how we can simplify difficult arguments using this machinery. There are also several other very nice applications that can be observed in Section 6 of [DS11].

The Larger Picture

We discussed two different models of higher categories here. There is no reason to stop at 2 though. There are many more models out there, such as complete Segal spaces and Segal categories. A nice review of all major models can be found in a survey paper by Bergner [Be10]. Thus it is a common practice to prove results in higher category theory by using the model that is most convenient and switching models around as the tools required in a proof change. But this might make us wonder, doesn't the existence of all these models cause a problem if there are too many non-trivial automorphisms? In order words, there is a chance that by moving through these models and coming back at the end we might end up with an object that is equivalent to the original one but has completely different properties.

Fortunately, this turns out not to be true. There is a very strong result by Toën [To05], that proves that every reasonable model for a higher category (in the sense that it is equivalent to the complete Segal space model structure), has only two automorphisms, the identity and the one that sends each higher category to the opposite category. Thus no equivalence can fundamentally change the objects we are working with and we are justified in switching models as we see fit.

References

[Be07] J.E. Bergner, *A model category structure on the category of simplicial categories.* Trans. Amer. Math. Soc. 359, 2007, 2043-2058.

[Be10] J.E. Bergner, *A survey of (∞,1)-categories*, The IMA Volumes in Mathematics and its Applications. Springer, 2010. pp. 69-83

[DS11] D. Dugger, D. I. Spivak, *Rigidification of Quasi-Categories*, Alg. Geom. Topol. 11 (2011) 225-261

[Lu09] J. Lurie, *Higher Topos Theory*, Annals of Mathematics Studies 170, Princeton University Press, Princeton, NJ, 2009, xviii+925 pp. A

[Pa17] A. Panagiotopoulos, *Review of Quasi-Categories* Unpublished Notes http://math.illinois.edu/ rasekh2/seminars/quasicat.pdf

[To05] B. Toën, *Vers une axiomatisation de la théorie des catégories supérieures*, K-Theory, 34 (2005), no. 3, 233-263.